Aanjaneya Pandey

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Profile Summary

Final-year Electronics & Communication Engineering student at **NIT Delhi**, pursuing a Minor in **AI & ML**. **Semifinalist, Flipkart GRiD 8.0** (AI Engineering Track) and **Leetcode Guardian** with a **2,603** contest rating (top 0.1%), 700+ DSA problems solved. Built **AI-powered full-stack applications** and **LLM-based systems** using **Python, C++, React.js, Node.js, LangChain, and FAISS**, backed by strong **DSA** and system design fundamentals, plus research experience in performance benchmarking.

Education

National Institute of Technology Delhi, New Delhi — **Expected May 2027**
B.Tech in Electronics & Communication Engineering (Minor in AI & ML) — *CGPA: **8.80/10***
Relevant Coursework: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks, AI/ML

Ewing Christian Public School, Prayagraj — **2022**
CBSE Senior Secondary — *91%*

Technical Skills

Languages: C/C++, Java, Python, JavaScript, TypeScript, SQL
Backend & Systems: Node.js, Express.js, FastAPI, REST APIs, WebSockets, RabbitMQ, JWT
Databases & Infra: PostgreSQL, MongoDB, Redis, SQLite, Qdrant, Prisma ORM, Docker, AWS, Azure
Frontend: React.js, Next.js, Redux Toolkit, Tailwind CSS
AI / ML: LangChain, RAG, OpenAI/Groq API, YOLOv8, NumPy, Pandas, Hugging Face, FAISS, LightGBM, XGBoost
Testing & CI/CD: GitHub Actions, Unit Testing, Integration Testing, Postman, Version Control (Git)

Work Experience

Software Research Intern — **June 2025 – July 2025**
Indian Institute of Technology (BHU), Varanasi — GitHub

- Built low-level communication drivers in **C** on **Zephyr RTOS**; developed a cross-platform benchmarking framework across **3 operating systems** for automated performance testing.
- Benchmarked LoRaWAN communication stack across **3+ hardware configurations**, improving network efficiency by **30%** via data-driven parameter tuning.
- Applied probabilistic modelling to evaluate communication reliability under varying network conditions, boosting system reliability by **20%**.

Projects

NextFlow | Next.js, TypeScript, React, Prisma, PostgreSQL, Redis, LangChain — GitHub | Live

1. Architected a visual AI workflow automation platform with a **React Flow** canvas supporting **4 reusable node types**, drag-and-drop construction, and execution tracking.
2. Designed a **DAG-based execution engine** resolving dependencies and running independent nodes in parallel, tracking status, duration, and errors per step.
3. Unified **5 LLM providers** — Groq, OpenAI, Gemini, Claude, and Ollama — through a single execution layer with **PostgreSQL + Prisma** persistence and **Docker** orchestration.

Rayvex | FastAPI, PostgreSQL, Redis, RabbitMQ, LangChain, Groq API, Next.js — GitHub

1. Developed **Rayvex**, an AI revenue-recovery agent with a deterministic state machine and a **12-rule policy engine** overriding LLM-proposed actions, built using **FastAPI, PostgreSQL, Redis**, and RabbitMQ.
2. Integrated a **LangChain** tool-calling agent (**10+ structured tools**) with schema-validated, expected-value-based decisions, verifying recovery only through independent payment-state confirmation before marking revenue recovered.
3. Implemented a benchmark comparing recovery performance against a naive-retry baseline on a **5,000+ case** synthetic dataset, validated by **200+ automated tests** across unit, integration, and end-to-end paths.

IntruSight | Python, FastAPI, React.js, YOLOv8, OpenCV, SQLite, WebSockets — GitHub

1. Engineered a real-time AI video analytics platform for **4 concurrent camera streams** with live monitoring, event playback, and analytics dashboards.
2. Combined a **4-stage detection pipeline** — MOG2 gating, **YOLOv8** classification, zone filtering, dwell-time verification — cutting false alarms by **91%** (**878** raw triggers to **78** confirmed events).
3. Deployed multi-threaded per-camera capture with **FastAPI WebSockets** for real-time delivery and **SQLite**-backed persistence.

Achievements & Highlights

- **Semifinalist**, **Flipkart GRiD 8.0** (AI Engineering Track), advancing through multiple national-level coding rounds.
- Advanced to **Round 2** of **Google BigCode Contest**, ranking among the **top 1,500** participants globally.
- Secured **Rank #1** in **NxtWave Code Combat**, outperforming **1000+ participants** nationally.
- Secured a perfect **10.0 SGPA** in first year of B.Tech, reflecting strong academic consistency.
- Solved **700+ DSA problems** across platforms; achieved **Guardian (2,603 rating)** on **LeetCode**.